

NIFTY 100 ANALYTICS

Analyst Guide

Operational guide for the Nifty 100 financial analytics system.

1. Project Overview

End-to-end Nifty 100 analytics platform covering ETL, SQLite storage, financial ratios, screening, peer analysis, valuation, cash-flow intelligence, NLP, clustering, reporting, Streamlit and FastAPI.

2. Repository Architecture

src/etl handles ingestion and validation. src/analytics contains financial analytics. src/screener contains screening logic. src/nlp contains parsing and pros/cons generation. src/reports creates PDF reports. src/dashboard contains the eight Streamlit screens. app contains the FastAPI service.

3. Database

The main database is db/nifty100.db. Company identifiers must be trimmed and converted to uppercase before joins. Monetary values are stored in INR Crore.

4. ETL

Use pandas read_excel(path, header=1) for core Excel files. The loader, normaliser and validator form the ETL pipeline.

5. Financial Ratios

The analytics engine calculates ROE, ROCE, debt-to-equity, operating margin, CAGR, cash-flow and related KPIs. Negative CAGR base years use TURNAROUND; zero interest expense uses Debt Free.

6. Screener and Peers

The screener supports configured financial thresholds and peer comparisons. The D/E filter skips the Financials sector.

7. Valuation

Valuation includes P/E, P/B, EV/EBITDA, FCF yield, five-year median P/E and sector-median comparison. Results are available through the valuation API.

8. Cash Flow and NLP

Cash-flow intelligence evaluates operating cash flow, free cash flow and capital allocation. NLP parses analysis data and generates structured company pros and cons.

9. Clustering

Companies are standardized and clustered using financial features. The current clustering output contains 92 companies and the API reports 12 outliers.

10. Reports

The project generates 92 company tearsheets, radar charts and sector/portfolio reports through src/reports.

11. Streamlit Dashboard

The dashboard has eight screens: Home, Profile, Screener, Peers, Trends, Sectors, Capital and Reports. The standard target is localhost:8501.

12. FastAPI

The API is implemented in app/main.py with company, financials, analytics, summary and valuation routes. The service runs on localhost:8000.

13. Testing

The current suite contains 65 tests. The latest verified execution passed all 65 tests with zero failures. The HTML report is reports/pytest_report.html.

14. Analyst Rules

Label stock_prices and market_cap as SIMULATED. Preserve INR Crore units. Normalize company_id before joins. Run tests before every Git commit.

15. Final Acceptance

All 23 deliverables must exist in their specified locations. Zero test failures are mandatory. Verify the API, dashboard, reports and Git repository before project sign-off.